ServiceNow Notes CSA

Dot-walking in the Now Platform is a term used to describe accessing fields on related tables from a form, list, or script.

ServiceNow Product Documentation - https://www.servicenow.com/docs/

My Work: list of all active tasks assigned to you, including:

- Change Request
- Group Approval
- Incident
- Knowledge Base Submission
- Request
- Security Case
- Visual Task Boards

My Groups Work: list of all active tasks assigned to your group(s) but not yet assigned to an individual.

Visual Task Boards (VTBs). Visual Task Boards allow you to:

- Manage tasks through a visual, drag-and-drop interface
- Identify process bottlenecks at a glance, in real-time
- Track embedded activity screens to view updates all in one place

<u>Change Homepage Color:</u>

System Properties > CSS

Installation exits - contains details for how a user will be redirected once he is logged out from the instance

Plugins - Data inconsistency correction is done from automated?

There are two types of VTBs: **Freeform** and **Data Driven**.

1. Freeform - Use Freeform boards as your personal organizer, creating individual tasks of any kind and freely adding, removing, and modifying cards and lanes. You may change the title of your lanes created in a Freeform board.

2. Data Driven - Allows you to add tasks to a flexible or guided task board. Create filters to display specific records from a table. You may not change the title of lanes presented in a Data Driven Board.

Visualizations from lists only show the current state of the data, Platform

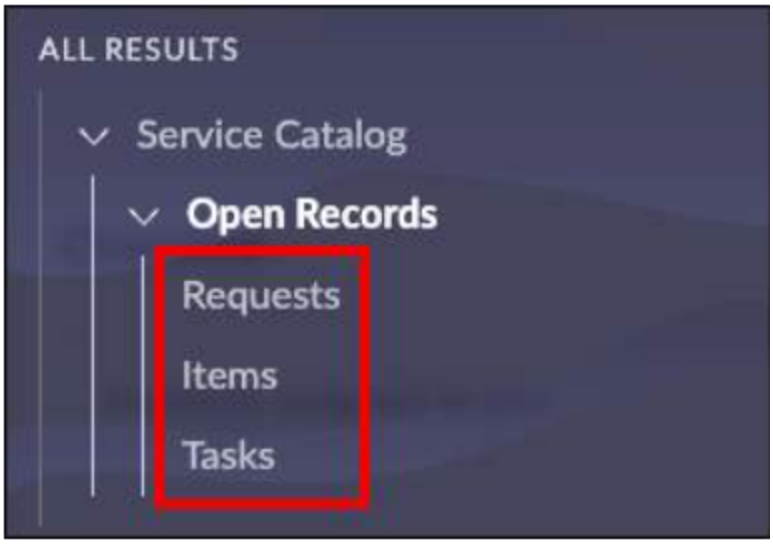
Analytics (PA) shows the data over time and is used for dashboards.

Administrators can create multiple **Knowledge Base**s and assign them to individual users in the Platform. This is called **User Criteria**. User Criteria defines conditions that are evaluated against users to determine which users can **create** (upload or author), **read** (view), **write** (edit), and **retire** (make articles inaccessible to all users) knowledge articles. User Criteria is applied at the Knowledge Base level in the Platform.

All > Knowledge > Administration > User Criteria

Service Catalog is a robust ordering system for services and products offered by various departments for users

- Once a catalog item (requests, requested items, and/or tasks) has been ordered, you may check on the status of the item by navigating to **All > Service Catalog > Open Records, and select Requests, Items, or Tasks.**



<u>Automation Assignment Enablement</u> - troubleshooting

2. Check the "Assignment Lookup" Rules

The most common way to fix this is via **Data Lookups**.

- Navigate to **System Policy > Rules > Data Lookup Definitions.**
- Look for the **Assignment Lookup** record.
- You likely need to add a matcher row where:
  - **Service Offering** = Strawberry sFone
  - **Assignment Group** = Strawberry Support

Assigning Roles:

> **Tip:** It is recommended to create logins for the following roles to effectively test the system:
>
> - admin – to do work
> - itil – to test as a process user or a similar role based on specific applications
> - ess (employee self service) – to test as an end user

Application > Module access	catalog role	user_criteria_admin role	catalog_admin role
Service Catalog	✓		✓
Service Catalog > User Criteria		✓	✓
Knowledge > User Criteria		✓	✓
Contextual Search			✓
Item Designer			✓
Service Creator			✓

How to secure and separate Sysadmin and HR Admin roles link:

https://www.servicenow.com/community/hrsd-articles/hr-security-setup-hr-admin/ta-p/2310649

What are some of the ways to interact with the ServiceNow Platform?:

- Workspaces
- Service Catalog
- Dashboards
- Lists
- Forms
- Knowledge Bases

What are some of the features of Employee Center?

- Easy to configure options
- Easy scalability
- Customizable portal pages
- Unified browser for all content types
- Built-in extensibility
- Integrated workflows
- Pre-enabled widgets
- Curated pages for popular topics
- Employee-centric navigation
- Intranet-like engagement and communication

Baseline implementation - is a set of installed apps on a ServiceNow instance, before any configuration or customization has been done.

Applications and plugins

	Applications	Plugins
Availability	Through the ServiceNow Store	Only available through the **All** menu. **All > Admin Center > Application Manager** or typing **v_plugin.list** into the Filter navigator
Origin	Developed by ServiceNow Partners or ServiceNow	Exclusively developed by ServiceNow
Scope	Usually shipped in a custom application scope	Can either be in a custom application scope or in 'Global'
Size	Usually larger bundles, lots of features	Ranges from tiny plugins containing a single dashboard to large plugins containing hundreds of files

Update browser tab label

Section 2: Update browser tab label

1. Navigate to **All > System Properties > System.**

Update logo image:

Update image on the My Company record

Marketing has provided a Cloud Dimensions logo, and IT has added DEV to the logo image.

1. Ensure you are logged in as **System Administrator**.

   Notice the ServiceNow **default logo**.

2. Navigate to **All > System Properties > My Company**.

Adding a new column named “Pilot” to incident table, use the “System Definition”

Section 1: Add Pilot column to incident table

In this section, we will add the new **Pilot** field you created to a new list view.

1. Ensure you are logged in as **System Administrator**.
2. Navigate to **All > System Definition > Tables**.
3. Search for the **Incident table**.

Formatters - an element used to display information that is not a field in the record for instance the activity stream (displays a list of activities)

- Activity formatter (journal fields like comments and work notes)

- Process flow - show different stages in a linear flow across the top of a record
- Parent breadcrumbs - provides crumbs to show the parent(s) of the task
- Approval summarizer - show dynamic summary info about req being approved
- CI Relations - shows on the CI form & allows for viewing of relationships between the current CI and related CIs.

How to:
To disable personalize form for the itil role, navigate to **sys_properties.list** find the property **glide.ui.personalize_form.role** and set the **Value** to **admin** (to specify the role that can access form personalization).

Changing Number Format per Table > **Number Maintenance**

Record numbers can be automatically incremented, and the number format per table in the system can be changed by visiting the **All > System Definition > Number Maintenance** application.

For example, the default problem record number prefix is "PRB" but can be redefined as "PRBLM."

Database Views: A database view defines table joins for reporting purposes. For example, a database view can join the Incident table to the Metric Definition and Metric Instance tables. This view can be used to report on incident metrics and may include fields from any of these three tables. The data in the virtual table created by a database view is read-only. Create Database Views by navigating to **System Definition > Database Views**.

Extensions: The extended table includes unique fields plus all of the fields and their properties from the parent table.

CRUD (Create, Read, Update, Delete) - several levels of security before an end user has the capability to perform these.

ACL - evaluates role, conditions, script
- A user must pass both table and field ACL rules to access a record object

Access controls are evaluated in the below order:

1. First at the Table-level (most specific to most general)
2. Then at the field-level (most specific to most general)

There are three security modules typically used by the System Administrator:

- **All > System Properties > Security**
- **All > System Security > Access Control (ACL)**
- **All > System Security > High Security Settings**

Additional ServiceNow operation examples include:

- **execute**: User cannot execute scripts on a record or UI page.
- **Edit_ci_relations**: User cannot define relationships between Configuration Item [cmdb_ci] tables.
- **Save_as_template**: Controls the field that should be saved when a template is created.
- **Report_on**: User cannot create reports on the object.
- **Personalize_choices**: User cannot right-click a choice list field and select Configure Choices.

Each Access Control rule specifies:

- A valid **operation** - a valid action the system can take (CRUD)
- The **object** being secured (e.g., table, table AND field)
- The **permissions** required to access the object:
  - Roles
  - Conditional Expressions
  - Scripts

Coalescing - a field means the field will be used as a unique key during imports. (Used to prevent duplicate records)

- Allows you to update existing Targeting Table records when importing data from an import set
- If a match is not found using the coalesce field, then a new record will be inserted into the database.

<u>CMDB</u>:
Access to the CMDB tables and underlying data requires certain permissions, such as the following roles:

- asset
- itil
- itil_admin
- cmdb_read

3 key CMDB tables are **Base Configuration Item [cmdb]**, **Configuration Item [cmdb_ci]**, which contains CI data, and **CI Relationship [cmdb_rel_ci]** which contains CI relationship data.

- cmdb

- cmdb_ci
- cmdb_rel_ci

CMDB Data Manager - https://www.servicenow.com/docs/r/zS6yb8e~gR8BPjeviYdV9g/6I4OMRcPHLSAka2kYguM7g

Plugins: The following plugins must be activated:
- CMDB CSDM Support (com.snc.cmdb.csdm)
- CSDM Activation (com.snc.cmdb.csdm.activation)

CS (Common Service) - a standard & shared set of service-related definitions across our products and platform that will enable and support true service-level reporting

DM (Data Model) - a CMDB Framework across our products and platform that will enable and support multiple configuration strategies.

CSDM (Common Service Data Model) - is a set of terms and definitions that can be used with all ServiceNow products on the Now Platform. Enables service reporting & provides prescriptive guidelines for service modeling within the CMDB. Used to map your IT services to ServiceNow.

CSDM: What it is, what is it NOT, and what's in it for you

What it is

- Standard terms and definitions
- Best Practice for CMDB Data Modeling and Data Management
- Out-of-the-box CMDB core tables
- Guidance on service modeling
- Recommended mappings

Knowledge Base

contribute to the knowledge base. If there is no user criteria on the Can Read related list, then the knowledge base becomes open to the public. To control access to logged in users only, administrators should leverage the **glide.knowman.block_access_with_no_user_criteria** property.

	admin	knowledge_admin	knowledge_manager	knowledge_coach	knowledge_domain_expert	knowledge_group_manager	knowledge_group_member	knowledge (can contribute)	Knowledge (can view)
Create knowledge base	•	•							
Configure workflows	•								
Modify knowledge base: categories, article validity, icon, workflow, description, pin articles, article approval	•	•	•						
Delete knowledge base									
Create and modify article templates	•	•							
Create, import, and/or modify article	•	•	•	•	•	•	•	•	
Enable article versioning, group ownership, multi-language search, actionable feedback, subscriptions	•								
Modify knowledge homepage, search result properties, subscription, AQI properties & surveys	•								
Execute AQI Surveys				•	•				
Manage knowledge group and feedback			•			•			
Approve knowledge, assign feedback to themselves							•	•	
View knowledge articles, provide feedback	•	•	•					•	•
Track usage									

> **Note:** If the publish checkbox is not visible, the system administrator will need to modify a system property. To show the **Publish** check box, navigate to **All > Knowledge > Administration > Properties** and activate the **Show publish checkbox on the knowledge import pop-up** property.

Service Catalog

Create a new service catalog item:

Section 2: Create new Service Catalog item

2. Navigate to **All > Service Catalog > Catalog Builder**.

To create a new item or modify an existing item, navigate to **All > Service Catalog > Catalog Definitions > Maintain Items.**

> **Note:** Use a record producer to create task-based records only. To ensure that standard service catalog processes are followed, such as initiating flows as expected, **do not create requested item records from record producers**. Instead, create requested item using catalog items.

Order Guides provide the ability to order multiple, related items as one request. Remember that variables are presented by the **Order** field number. Use an Order Guide to assist users in determining what items they need.

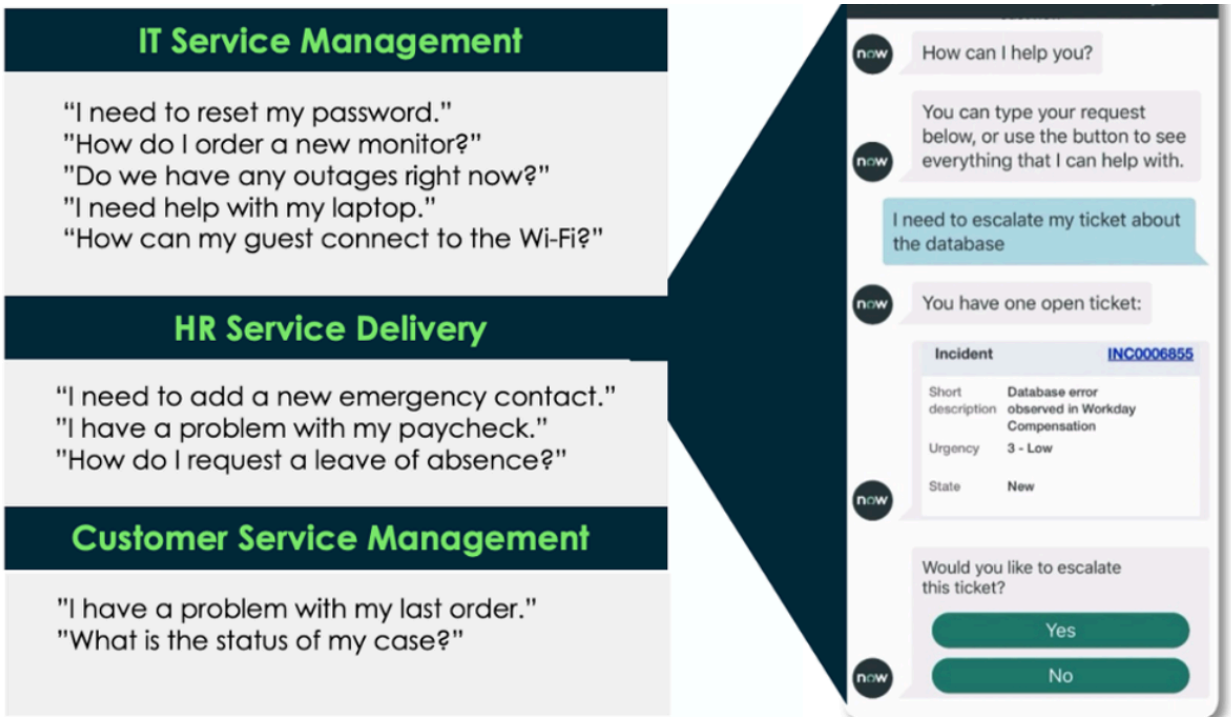
User Criteria - defines who can and cannot access KB articles.
Variables - represents questions asked when ordering a catalog item (color, storage, etc.)

What is created when an order is placed for a catalog item?:

1. **REQ#**
2. **RITM#**
3. **SCTASK#**

<u>Virtual Agent</u>

Come request that are asked:



When creating a catalog item and it get stocks in review state apply this update set to resolve the issue:
https://servicenow.sharepoint.com/sites/CDevExternal/ExtInstructorShare/Forms/AllItems.aspx?id=%2Fsites%2FCDevExternal%2FExtInstructorShare%2F

ServiceNow%20Administration%20Fundamentals%20Instructor%20Share%2FXanadu%2FInstructor%20Share%2FServiceNow%20Administration%20Fundamentals%20Course%20Materials%20%28English%29%2Fcatalog%5Fbuilder%5Fupdate%5Fset%5F7dfc68319fd212107f443e392024ab5d%2Exml&parent=%2Fsites%2FCDevExternal%2FExtInstructorShare%2FServiceNow%20Administration%20Fundamentals%20Instructor%20Share%2FXanadu%2FInstructor%20Share%2FServiceNow%20Administration%20Fundamentals%20Course%20Materials%20%28English%29&p=true&ga=1

Workflow Studio:

Spokes - “connectors” that allow you to talk to 3rd-party software w/o having to write custom script from scratch

Playbooks:

With Playbooks, you gain these benefits:

- Connect multiple flows and actions into an end-to-end business workflow.
- Reuse existing flows, subflows, or actions to automate process activities.
- Organize process activities in a digitized task board interface.
- Guide end users to complete a process in a task-oriented interface, such as a playbook.
- Consolidate separate business processes across the organization.
- Define a consistent record life cycle from creation to completion.
- Pass data between the activities and stages of a business process.
- Specify the conditions and the order in which activities and stages run.

More info on playbooks:
https://www.servicenow.com/docs/r/0MvJg~UYByX0VxWLomI71w/qBJsjRiKdwxYUWrBTaIgUw

When creating a **NEW FLOW** from scratch: **VERY IMPORTANT**

> **Note:** The Run as option specifies if the flow runs as system user or the user who initiates the session. Be sure to understand the differences between the two options. Select the user who initiates the session option when updates should come from the user who triggered the flow. System user is a service account that is not restricted by access controls.

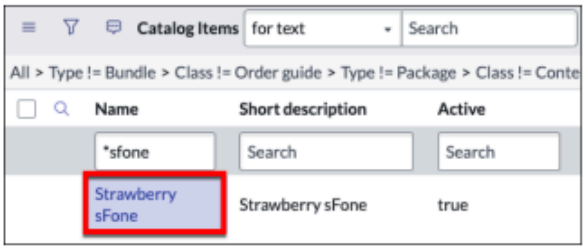
To associate a flow with an item **Maintain Items > Locate item record > Scroll to Process Engine > Add workflow name in the 'Flow' field > Select update**.

L. Activate and associate flow to Service Catalog Item

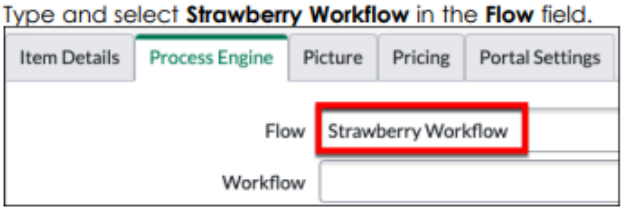
1. Select the **Activate** button.

> **Note:** *A message will appear confirming you would like to activate the flow – select* ***Activate****.*

2. Return to the main ServiceNow window and navigate to **All > Service Catalog > Catalog Definitions > Maintain Items**.
3. Locate and open the **Strawberry sFone** item record.
4. Scroll down and select the **Process Engine** tab.



5. Type and select **Strawberry Workflow** in the **Flow** field.



6. Select **Update** to save the Strawberry item record and return to your previous screen.

1. Navigate to System Properties by typing **sys_properties.list** in the Filter Navigator.

2. Locate the **com.snc.process_flow.reporting.level** and ensure the record is saved with the **ON** value. You can do this by opening the record and typing the value ON, or you can edit directly inline by double-clicking in the field.

> **Note:** *This is required to view the Flow execution at the end of the lab verification and troubleshoot if necessary*

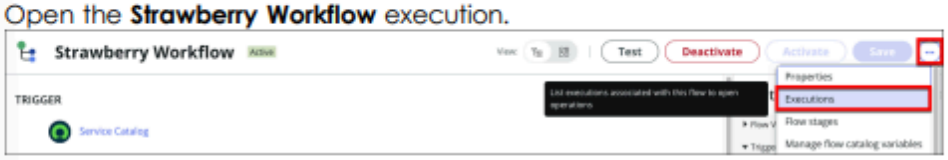
HOW TO CHECK FLOW EXECUTION IN WORKFLOW STUDI0

C. Review Flow Execution

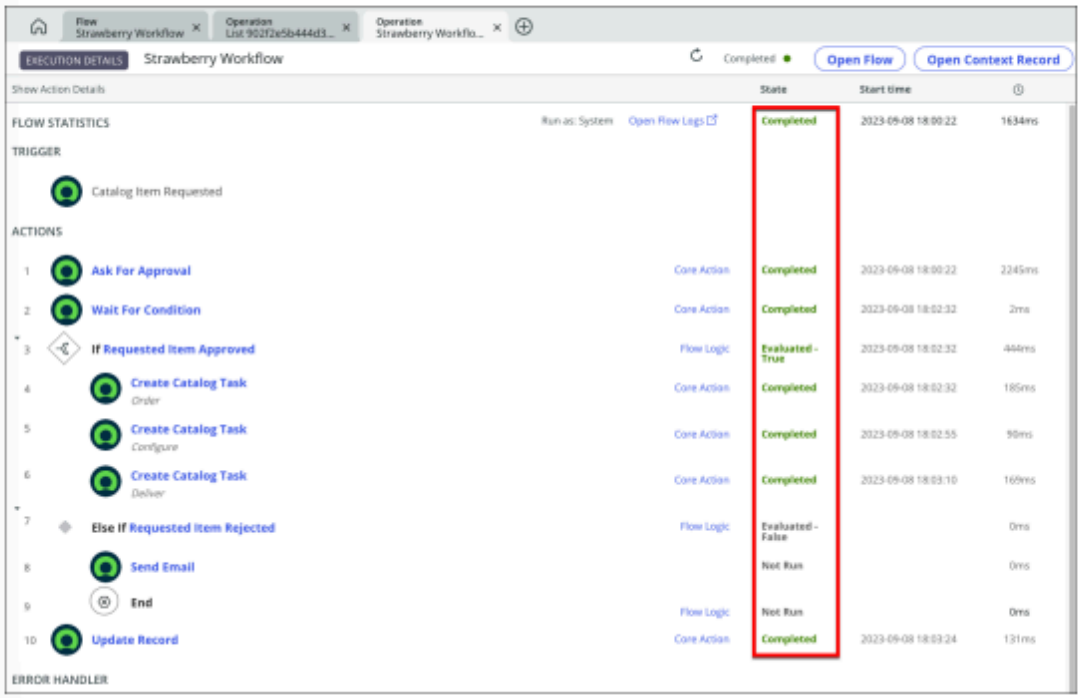
1. Navigate to **All > Process Automation > Workflow Studio**.

2. Locate, then select the **Strawberry Workflow**.

3. Open the **Strawberry Workflow** execution.



4. Open the **Strawberry Worflow** and review the status of the workflow (**Completed**) and tl State of each step. Is it what you expected? If not, troubleshoot and work through the Lab Validation steps again.



Data from an action is stored in a “**data pil**l” so it can be reused.

Components of a flow:

1. Trigger
2. One or more actions

<u>Flow Designer Triggers:</u>

Triggers

Triggers can be *record-based, date-based,* or *application-based*.

- **Record-based** triggers run a flow after a record has been created or updated.
- **Schedule-based** triggers run a flow at the specified date and time or at scheduled intervals.
- **Application-based** triggers start a flow when application-specific conditions are met.

IMPORTANT TO KNOW:

- The ServiceNow platform comes with several baseline portals and self-service options.
- Knowledge management allows users to create, categorize, review, approve, and browse information in a centralized location.
- Service Catalog provides a "one-stop-shop" for users to order services and products.
- Catalog Builder is designed to support common and simple use cases for business users.
- Workflow Studio is a non-technical interface for building and enabling process automation.
- Playbooks in Workflow Studio is targeted for developers and process owners. It enables owners to author cross-enterprise workflows and create a single unified process.

Platform Analytics:

If none of existing visualizations meet your needs, you can create your own by navigating to **All > Platform Analytics > Library > Data Visualizations (or All > Platform Analytics > Analytics Center > Create new visualizations)**. Alternatively, you can simply select most column context menus in any list to generate a chart directly from the data in that list.

For more information, navigate to **All > Platform Analytics > Analytics Center.**

Performance Analytics can generate accurate historical trends by capturing continuous snapshots on a regular schedule. As illustrated here, ad-hoc reporting answers the question of "Where are we today?" while performance analytics answers what is happening over time.

In addition to calculating trends from snapshots, Performance Analytics can:

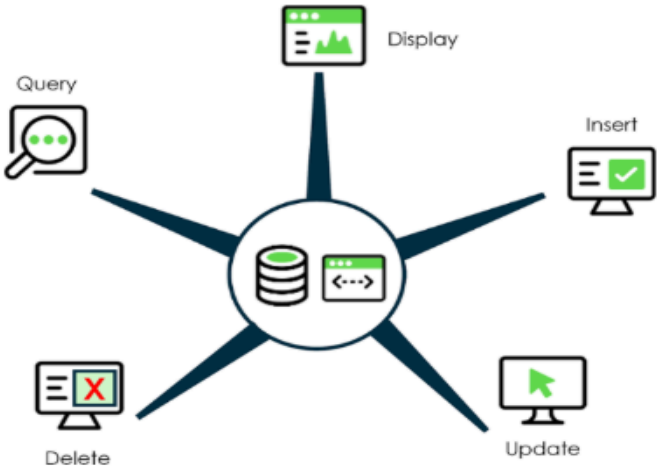
- Track performance against Targets
- Alert when Thresholds are met
- Forecast future performance
- Compare performance at different points in time

When sharing visualizations and dashboards with colleagues you have to grant access to **BOTH**

> **Note:** To share visualizations and dashboards with colleagues, you will need to grant access to **share the visualization AND dashboard** to each group or person. If you grant access to a dashboard alone, viewers will not be able to see the visualization unless they have access to both.

Business Rule:

A **Business Rule** is configured to run when a record is displayed, inserted, updated, deleted, or when a table is queried.



Email Notifications

Email notifications are used to send selected users email or SMS notifications about specific activities in the system, such as updates to incidents or change requests. To access a new notification record, navigate to **All > System Notification > Email > Notifications**.

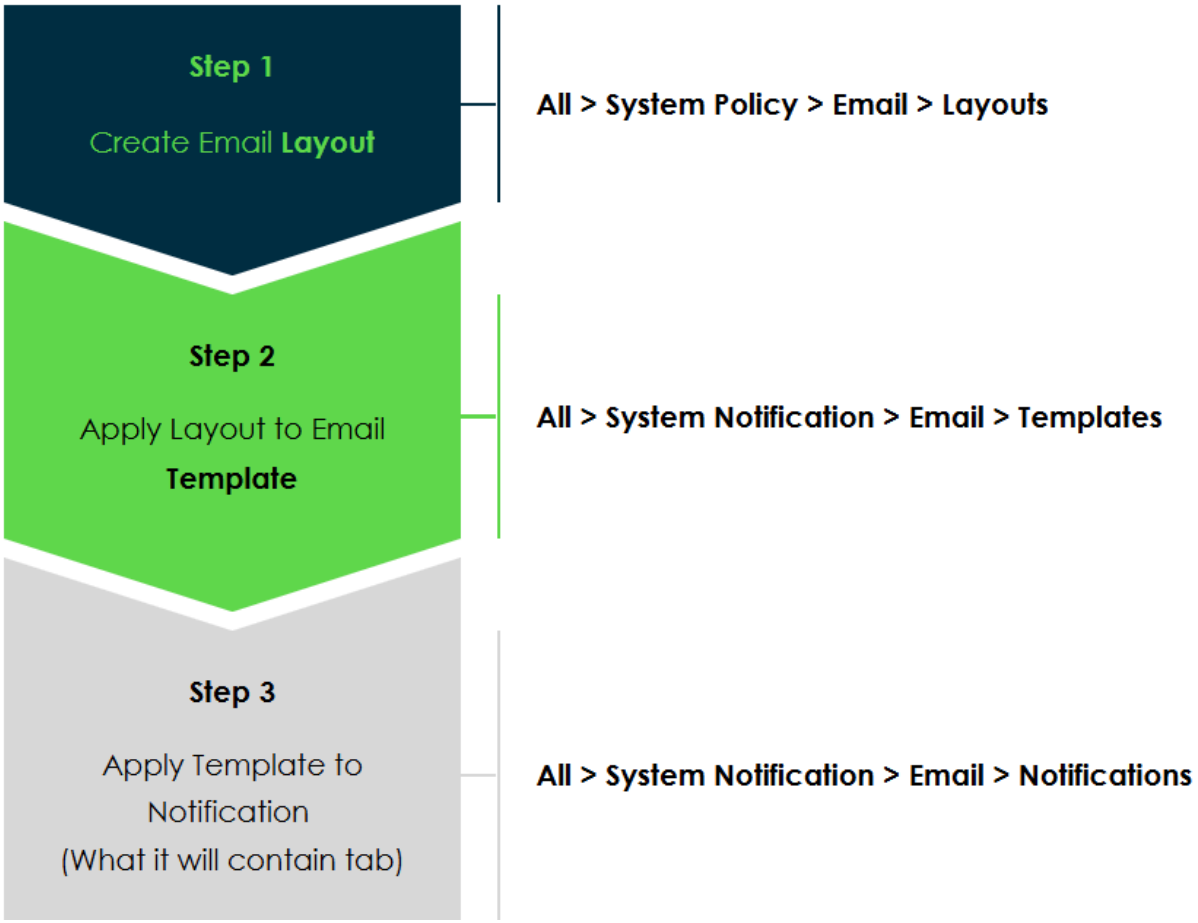
> **Tip:** To view notifications in your instance, navigate to **All >System Mailboxes > Outbound > Outbox**. Right-click on Created Date and select Preview Email.

Tips	Pitfalls
• The baseline system provides several email notifications. • Explore editing an existing notification before creating a new one.	• Customers tend to overdo e-mail notifications. • Customers get overwhelmed and confused by the number of different notifications. • Customers specify requirements for caller-facing notifications too late into implementation.

> **Tip:** Consider limiting the recipient list of any notification to 1000 users. By default, if a notification has more than 100 intended recipients, the system creates multiple notification messages with up to 100 recipients each. If you want to change the recipient limit, set the system property **glide.email.smtp.max_recipients.**

If using an **Email Template,** then **Subject** and **Message** will be used from the template unless overridden with a different (new) subject and/or message.

Using email layouts and templates with notifications



> **Note:** The ${URI_REF} placeholder renders as a direct link to the incident record. If needed, you could select the link to review the incident in a separate tab.

Questions to study:

1. **What should you consider when exporting a visualization to PDF and sharing the exported file?**

Hide

- Information may be obsolete
- Access control is not enforced on exported data

2. **What are some configuration options in the Visualization Designer?**

Hide

- Data, Type, Metrics, and Presentation

3. **What can trigger a notification?**

Hide

- Record inserted or updated
- Event is fired
- A script

4. **Name three options to share or create copies of the visualizations?**

Hide

- Duplicate
- Share
- Export

5. **Which three tabs are used to configure an email notification?**

Hide

- When to send
- Who will receive
- What it will contain

6. **Where can users define notification channels and manage subscriptions?**

Hide

- All > Self-Service > My Notification Preferences ***(or)*** User profile menu > Preferences > Notifications > Core UI

UI Policies

To create UI Policies:

> **Note:** Alternatively, you could go to **All > System UI > UI Policies** and select the table name.

Update Sets - a group of configuration and customization changes that can be moved from one instance to another.

1. XML file
2. A set of record details that uniquely identify the update set
3. A list of configuration changes
4. State to determine if instance can retrieve & apply config changes

An Update Set is a container for configuration records. By navigating to **All > System Update Sets > Local Update Sets**, you can create a new Update Set or set an existing one as your current Update Set. Use an Update Set to migrate your code.

Creating an update set:
https://www.servicenow.com/docs/r/guL1zqmJkV9eywMq_6qu~w/8RuhnDayBg4zNVPw_aDnxQ

> **Note:** It is important to make the update set Current, as your future changes will be captured in the Current update set.

Retrieve an update set from a remote instance

In the **Production** instance:

1. Navigate to **All > System Update Sets > Update Sources**
2. Select the development instance with the completed update set you want to commit

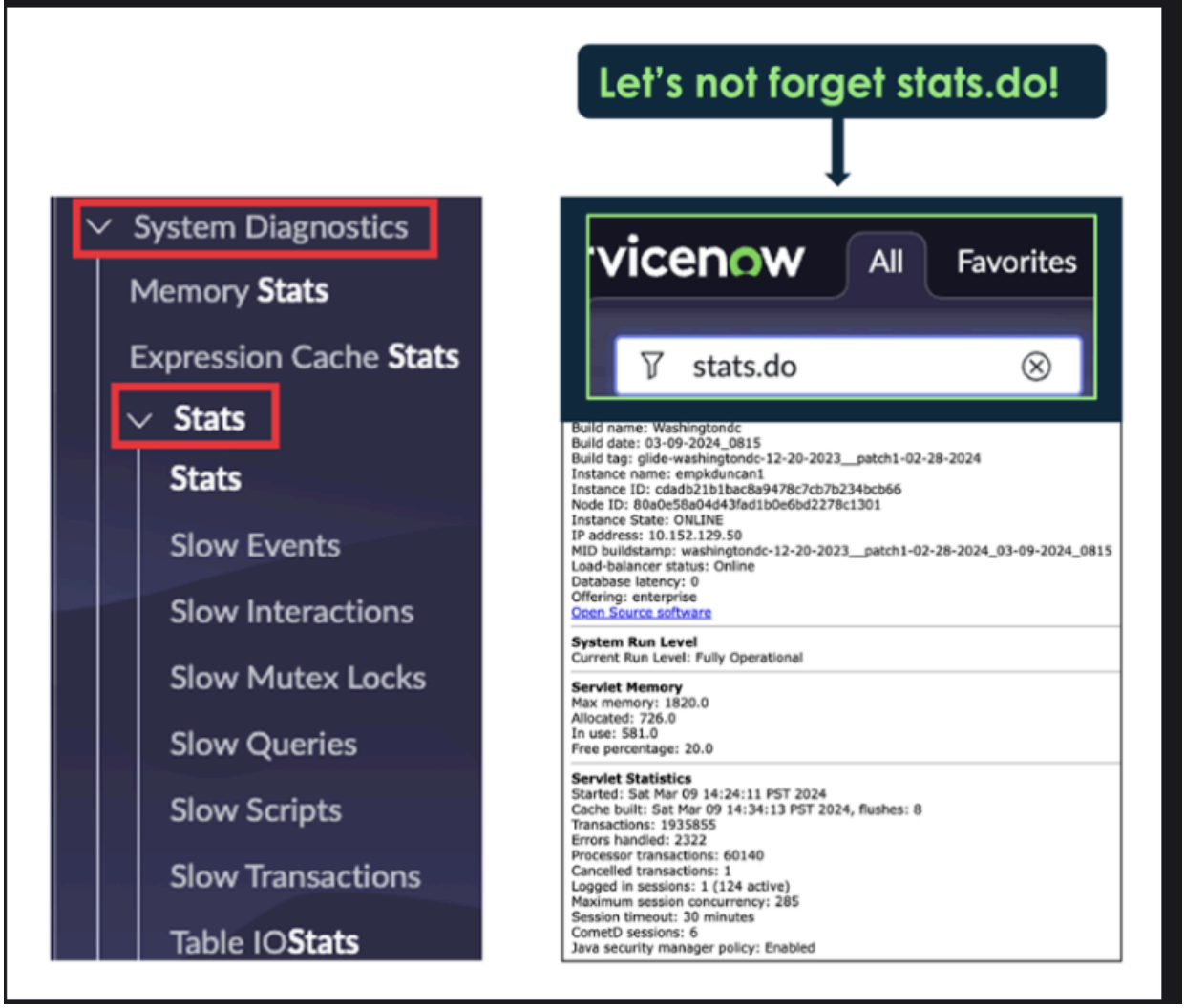
> **Note:** The first time you connect to a remote instance, you will need to specify the connections setting as described on the form (including Name, Type, Active, URL, Username, Password, and Short Description). Select **Test Connection** to ensure a successful retrieval. Save the new remote instance record.

Add'l resources:

Instance scan/HealthScan:
https://nowlearning.servicenow.com/lxp/en/now-platform/introduction-to-servicenow-healthscan-and-instance-scan?id=learning_course_prev&course_id=78370c2e47e8ca1090542034846d4336

System diagnostics stats tools:

To aid in performance evaluation, **Stats Tools** records statistics for system activities that affect performance, such as the execution of queries, scripts, and transactions.



Running release:
System administrators can check which release is running on an instance with **stats.do**

App Engine Studio (AES) - low-code development, automating business processes and solving business problems - allow you to delegate dvlp work that was once assigned to admins with little to no training.

Using AES, you can also:

- Create applications from scratch
- Access a list of recent applications you have built
- Access links to add objects to apps, browse app templates, and learn more about the available tools
- Browse through a list of templates you can use to create an app with preconfigured data, experience, automation, and security

Upgrades & Releases:

Stats.do - will tell you what release you are on
System logs>events>system.upgraded

Upgrading your instance involves planning, testing, and validation. Release notes guide you through completing the phases and tasks for a successful upgrade.

http://docs.servicenow.com/

https://www.youtube.com/watch?v=VJAFW6djQgU (Using the release notes)

To verify the upgrade is complete:

- **Navigate to All > System Diagnostics > Upgrade Log**
- Locate the message "Notifying HI that upgrade has been completed."

To confirm that a patch has been applied, find the system.patched event. You can create a notification for this event; it is not available by default.

Shared Responsibility Model

	Responsibility		
Area of Responsibility	Customer	ServiceNow	Colocation (Data center providers)
Secure configuration of instance	●		
Authentication and authorization	●		
Data management (classification and retention)	●		
Data encryption at rest	●		
Data encryption in transit	●	●	
Encryption key management	●	●	
Security logging and monitoring	●	●	
Secure SDLC processes	●	●	
Penetration testing	●	●	
Vulnerability management	●	●	
Privacy	●	●	
Compliance: regulatory and legal	●	●	●
Employee vetting or screening	●	●	●
Physical security/environment controls	●	●	●
Cloud infrastructure security management		●	
Infrastructure management		●	
Media disposal and destruction		●	
Backup and restore		●	
Business continuity and disaster recovery		●	

Logging & Monitoring:

Customers have access to various security-related information, including:

- Customer end-user activity
- Failed login attempts
- Audit trails and record histories
- Administrative escalation
- Access by ServiceNow support personnel

Maintaining a security contact - (how to)
https://support.servicenow.com/kb?id=kb_article_view&sysparm_article=KB0547262

Visit ServiceNow Security Center for help with securing the platform.

SC resource link:
https://support.servicenow.com/kb?id=kb_article_view&sysparm_article=KB0756556

Security Center updates 2x a year, check resources for how to update.

Workflow Types the Now Platform provides:

IT Workflows: Includes ITSM (Incident, Change), ITOM (Operations), and Strategic Portfolio Management.

Employee Workflows: Focused on the employee experience, primarily through HR Service Delivery and Legal Service Delivery.

Customer Workflows: Designed for external-facing service, including Customer Service Management (CSM) and Field Service Management.

Creator Workflows: The tools used to build custom apps and integrations, powered by App Engine and Integration Hub.

Guided setup > ITSM setup - helps guide setups

5. Where can users define notification channels and manage subscriptions?

- All > Self-Service > My Notification Preferences ***(or)*** User profile menu > Preferences > Notifications > Core UI